This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Encoding Data-Driven Flexible and Safe Regions in Distribution Network Operation via Constraint Learning

Linwei Sang, Zaijun Wu, *Senior Member, IEEE*, Yinliang Xu, *Senior Member, IEEE*, Mingyang Zhang, Qinran Hu, *Senior Member, IEEE*, and Hongbin Sun, *Fellow, IEEE*

***Abstract*—The decarbonization of distribution networks poses operational challenges with renewable uncertainty, and the attached digitalization provides emerging chances with data for complex flexible and safe region formulation. However, data-driven regions cannot be utilized for distribution network operation directly. This article proposes to encode the data-driven flexible and safe regions in distribution network operation. It formulates the identification of data-driven flexible and safe regions as binary classification tasks. The sparse neural network is utilized for classifying based on its classification performance for region learning and fewer parameters for network optimization. The trained region classifiers are transformed into a series of mixed-integer linear constraints via the big-M relaxation method to formulate the data-driven region constraints. The case study verifies first, the classification accuracy of region learning and second, the economy and empirical safety of distribution network operation with data-driven regions.**

***Index Terms*—Distribution network operation, flexible region, safe region, sparse neural networks (SNNs).**

I. Introduction

TO MITIGATE climate change, emerging renewables and flexible demand are penetrated in the active distribution networks for decarbonization [1], which poses significant operational challenges. Driven by the digitalization of distribution networks, advanced sensing, communication, and computing capabilities are increasingly being deployed, enabling data-rich and computationally efficient distribution network management. [2], [3], [4], [5], [6]. Both flexible and safe regions become complex and varying due to increasing distributed energy resources. Data-driven flexible and safe regions could be possible solution [7], [8].

Under the above transformation of decarbonization and digitalization, the distribution networks are becoming complex with enhanced controllability for the safe guarantee [6]. From the load demand perspective, its flexible region facilitates the exploitation of regulation capability, and from the system operator perspective, the safe region characterizes the secure operating space of the system. However, the data-driven regions could not be utilized for distribution network operation directly. So this article proposes to formulate the data-driven regions with neural networks (NNs) and encode them in distribution network operation in the way of fusing classification and optimization.

To facilitate the distribution network decarbonization, flexibility regions are constructed in a model-driven way via the optimization techniques [9], [10], [11], [12] and in a data-driven way via the learning techniques [7], [8]. For the model-driven flexibility region, the aggregate flexibility of virtual power plants is formulated via a novel high-dimensional polytope in [6]. An improved inner approximation for aggregating power flexibility is proposed in [11] and further incorporated into the security-constrained unit commitment. Two-stage robust optimization is utilized in [12] to achieve power flexibility aggregation at substations. The above model-driven approaches approximate the flexible region explicitly in a conservative way, which could be applied for system optimization directly. However, the above corresponding region formulation relies on detailed resource parameters, whose shapes are fixed and presumed.

So the data-driven approaches investigate to characterize the flexible regions by labeling the flexible data and learning the flexible region [7], [8]. A data-driven aggregate flexibility modeling is designed in [8] via a convex quadratic classifier. A network-informed data-driven approach is proposed in [7] to train a NN-based classifier to learn the feasible power region with efficient data sampling. The above data-driven approaches approximate the flexible region with arbitrary shapes by training with labeled data in a data-driven way. However, the trained NN-based classifier of [7] could not be utilized in downstream system optimization. The safe regions of distribution networks could also be formulated in data-driven and model-driven ways.

Under the digitalization of distribution networks, advanced artificial intelligence could further revolutionize smart grid

Received 10 February 2026; revised 21 March 2026 and 14 June 2026; accepted 15 August 2026. This work was supported by the National Natural Science Foundation of China under Grant 52507085 and Grant U24B2081. Paper no. TII-26-1444. *(Corresponding author: Yinliang Xu.)*

Linwei Sang, Zaijun Wu, and Qinran Hu are with the School of Electrical Engineering, Southeast University, Nanjing 210096, China.

Yinliang Xu is with the Tsinghua Shenzhen International Graduate School, Tsinghua University, Shenzhen 518055, China (e-mail: xu.yinliang@sz.tsinghua.edu.cn).

Mingyang Zhang is with the School of Electrical and Electronic Engineering, North China Electric Power University, Beijing 102206, China.

Hongbin Sun is with the Department of Electrical Engineering, Tsinghua University, Beijing 100084, China.

Digital Object Identifier 10.1109/TII.2026.3725029

1941-0050 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

system operation and control [13], [14], [15], [16], [17], [18]. In [13], the trained machine-learning surrogate can map power injections to voltages and branch flows in milliseconds, enabling fast approximate power-flow evaluation without solving the full AC power-flow equations each time. Zhang et al. [14] reviewed the data-driven security and stability rule extraction with learning under renewable penetrated power system operation. Constraint learning approaches are proposed in [19] and applied in power systems for data-driven network management in [15] and carbon-aware energy management in [16] with NNs. Learning with advanced NNs offers high representational capability, but suffers from limited interpretability due to their implicit neuron structures. In contrast, optimization models provide explicit formulations and clear interpretability, yet their reliance on predefined mathematical forms restricts representational flexibility. This article seeks to integrate the strengths of both approaches, bridging the gap between learning and optimization to achieve economically efficient and safe operation of distribution networks.

To utilize data-driven flexible and safe regions in distribution network decision-making, this article proposes to formulate the data-driven flexible and safe regions with NN-based region classifiers and encode the data-driven flexible and safe regions as constraints in distribution network operation, which bridges the gap between data, models, and decisions of distribution network operation. The identification of data-driven flexible and safe regions from offline sampled data is formulated as binary classification tasks. Various learning models could be constructed as region classifiers and trained for the above classification tasks with flexible and safe region datasets. Specifically, sparse neural networks (SNNs) are utilized based on the classification performance and fewer parameters. The trained data-driven region classifiers are transformed into a series of mixed-integer linear constraints for data-driven regions, which are encoded in distribution networks for data-driven economic and safe operation. So the main contribution of this article can be summarized as follows.

1) This article, *for the first time*, formulates the data-driven flexible and safe region identification as classification tasks and trains the learning-based classifiers to construct data-driven flexible and safe regions for distribution network management. Compared to [7] with only flexible region identification and [8] with only flexible region modeling, this work extends the region identification framework to include both flexible and safe regions, enabling comprehensive operational decision-making from both load aggregator, and system operator perspectives.
2) Based on the data-driven regions, the trained flexible and safe region classifiers are equivalently transformed into mixed-integer linear constraints and encoded in the distribution network optimization. Compared to [7] whose trained NN classifier remains a opaque predictor without constraint transformation, the proposed approach systematically transforms trained classifiers into optimization-ready MILP constraints via big-M relaxation, enabling direct integration into distribution network operation models.
3) Considering both classification and optimization performance, the SNNs are constructed for the flexible and safe region classifiers with both higher accuracy and fewer parameters compared to conventional logistic regression (LR) and support vector machine models. Compared to [8] with a convex quadratic classifier of limited expressiveness, the proposed SNNs achieve higher classification accuracy while maintaining parameter sparsity.

TABLE I
FEATURE COMPARISON WITH CLOSELY RELATED WORKS

Feature	[8]	[7]	[19]	Ours
Flexible region	✓	✓		✓
Safe region				✓
NN-based classifier		✓		✓
Sparse-NN classifier				✓
Classifier → MILP			✓	✓
Encoding in dispatch	✓		✓	✓
Power-system scope	✓	✓		✓

Novelty and Contribution Declaration: This article is a constraint learning application article for distribution network management, inspired by the general constraint learning framework in [19]. Concretely, it differs from [19] in three key aspects. First, the scope is distinct: Maragno et al. [19] presented a general operations research framework for data-driven decision-making, whereas this work targets power system applications by encoding flexible and safe regions in distribution network operation. Second, the region construction differs: Maragno et al. [19] defined trust regions using the convex hull of observations to avoid extrapolation, while this manuscript derives flexible and safe regions through transformations of trained classifiers. Third, the classifier transformation approach varies: Maragno et al. [19] converted general classifiers into mixed-integer linear constraints, while this study leverages SNNs for efficient identification of flexible and safe regions. Similar constraint learning applications in power systems include [20], [21] and [22].

Methodological advance beyond an application pipeline: Beyond integrating existing components, the contribution rests on three methodological elements summarized in Table I. 1) *Unified region encoding:* the load-aggregator flexible region and the operator safe region are cast within a *single* classification-to-MILP formulation, whereas prior power-system constraint-learning works [7], [8] address neither both perspectives nor the safe region. 2) *Sparsity–tractability co-design:* sparsity is exploited not for inference speed but to reduce the number of big-M/binary ReLU variables in the embedded MILP, explicitly linking the learned architecture to the optimization complexity (see Section V). 3) *Constraint-replacement mechanism:* explicit network-security constraints are *replaced* by the learned safe-region constraint in (17), enabling parameter-free and privacy-preserving secure dispatch.

II. PRELIMINARIES FOR FLEXIBLE AND SAFE REGIONS FORMULATION

This section proposes to explore and exploit flexible and safe regions of the distribution networks for economic and safe operation from the perspective of learning-based, data-driven, and classification ways. The distributed energy resources are firstly formulated in Section II-A. The flexible and safe region

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2020 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

of system operation is formulated in Section II-D1 and II-D2. The region identification strategies for flexible and safe regions are proposed in Section II-D for generating the flexible and safe datasets, where data are the basis of learning algorithms.

A. Unified Flexibility Operation Model

Based on the above with [6], we formulate the uniform flexibility operation model with battery-like and generation-like models in as follows:

$$p_{d,t,\min} \le p_{d,t} \le p_{d,t,\max} \qquad \forall d \in \mathcal{D}\, \forall t \in \mathcal{T} \tag{1a}$$

$$e_{d,t} = e_{d,t-1} + \eta_d p_{d,t} \Delta t \qquad \forall d \in \mathcal{D}\, \forall t \in \mathcal{T} \tag{1b}$$

$$e_{d,t,\min} \le e_{d,t} \le e_{d,t,\max} \qquad \forall d \in \mathcal{D}\, \forall t \in \mathcal{T} \tag{1c}$$

where $p_{d,t}$ is the unified power demand of d in time t, which represents $p_{i,t}^{\mathrm{EV}}$ for electrical vehicles and $p_{i,t}^{\mathrm{cha}}/p_{i,t}^{\mathrm{dis}}$ for energy storage; η_d is the energy transition efficiency; $e_{d,t}$ is the energy state; $p_{d,t,\min}$ and $p_{d,t,\max}$ are the minimum and maximum power limits; $e_{d,t,\min}$ and $e_{d,t,\max}$ are the minimum and maximum energy state limit; $\mathcal{D}$ is the set of unified demand resources. The above unified operation models and its aggregation are constructed in a linear programming (LP) form. For real-world aggregator portfolios, there may exist the challenges of model heterogeneity with nonconvex structures, uncertainty and behavior variability, scalability and data requirements, and parameter identification and privacy. These challenges will change the data sampling process, but the generated data could also be learned by proposed region identifiers.

B. Flexible Region of Load Aggregator

From the perspective of the *load aggregator*, flexible region describes the set of all feasible load in the scheduling time, and the feasible region of massive distributed energy resources should be constructed to facilitate the distribution network operation. Based on the [8] and [23], the flexible region $\mathcal{S}_{\mathrm{flex}}$ could be formulated mathematically based on unified flexible operation models in the following:

$$\mathcal{S}_{\mathrm{flex}} = \{p_{\mathrm{ref}} | W_f p_0 \le z, p_0 = A_f p_{\mathrm{ref}} + b_f\} \tag{2}$$

where p_{ref} is the aggregate power; p_0 is the distributed energy resources power of different resources; $\mathcal{S}_{\mathrm{flex}}$ constructs the total flexible region; W_f, A_f, and b_f are the corresponding operation constraints, and detailed illustration could refer to [8] and [23] for clear delivery. The above formulation relies on the detailed parameters of all distribution energy resources, however, the actual parameters are unknown to the load aggregator. So this article proposes to learn the data-driven flexible region for load aggregators. The selection and grouping of load aggregators will affect the boundaries of flexible regions from the resource mix, portfolio diversity, and grouping strategy.

C. Safe Region of Distribution Network Operation

From the perspective of the *distribution system operator*, safe region describes the set of all feasible bus power in each time, and the safe region of various bus active and reactive power injections should be constructed to facilitate the distribution network's safe operation. The safe region $\mathcal{S}_{\mathrm{safe}}$ could be formulated mathematically based on the distribution power flow models to map from bus power injection to bus operation states as follows:

$$\mathcal{S}_{\mathrm{safe}} = \{(p_{inj}, q_{inj}) | W_s s \le z, s = A_s^p p_{inj} + A_s^q q_{inj} + b_s\} \tag{3}$$

where (p_{inj}, q_{inj}) is the bus injection power; s is the operation states of the distribution network, including the bus voltages and branch power flow; z limits the operation space; W_s, A_s^p, A_s^q, and b_s are distribution network constraints, and detailed illustration could refer to [6]. The above formulation relies on the detailed parameters of distribution network systems, however, the actual parameters could be unknown to the distribution system operator. Similarly, this article proposes to learn the data-driven safe region.

Based on the above, learning the flexible and safe region in a data-driven way could be treated as a classification problem to identify whether p_{ref} of (2) is flexible or not and whether $(p_{\mathrm{inj}}, q_{\mathrm{inj}})$ of (3) is safe or not. We note that the distribution network power flow, flexible region (2), and safe region (3) are coupled through the bus power injection intensely. i) The power flow will not determine the region shape of flexible regions but will change the power output from flexible regions; ii) The power flow are determined by bus power injections, which are also limited by safe regions.

In summary, the flexible region $\mathcal{S}_{\mathrm{flex}}$ is the set of all aggregate power trajectories (or aggregate power values) that can be feasibly disaggregated into individual flexible resources while satisfying their operational constraints (power/energy limits, intertemporal coupling, etc.). Mathematically, $\mathcal{S}_{\mathrm{flex}}$ is characterized by the feasibility of the flexible-region identification problem (i.e., whether a given p_{ref} admits a feasible disaggregation under the unified flexibility model). The safe region $\mathcal{S}_{\mathrm{safe}}$ is the set of bus injection powers $(p_{\mathrm{inj}}, q_{\mathrm{inj}})$ such that the resulting network operating state (voltages and branch flows) satisfies all security constraints (e.g., voltage magnitude limits and branch thermal limits) under the distribution power-flow model.

Remark 1. (Difference between classification and regression): The learning-based identification of flexible and safe regions belongs to the classification tasks with discrete outputs. In comparison with [16], the learning-based approximation of complex relationships in distribution networks belongs to the regression tasks with continuous outputs.

The above remark aims to clarify why our *region identification* problem is naturally formulated as a *classification* task rather than a regression task. The reason is that the labels are generated using indicator functions $\mathbb{I}_{\mathrm{flex}}[d_{\mathrm{ref}} = 0]$ in (4) and $\mathbb{I}_{\mathrm{safe}}[W_s s \le z]$ in (5), leading to binary labels, which is also aligned with the classifier formulation in [7].

D. Flexible and Safe Region Identification

Based on the above flexible and safe region formulation, we need to label the feasible/infeasible and safe/unsafe data points for both flexible and safe regions to provide offline flexible region dataset $\mathcal{D}_{\mathrm{flex}}$ and safe region dataset $\mathcal{D}_{\mathrm{safe}}$ for the corresponding flexible region learning and safe region learning,

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

where data is the basis of model training. So this part constructs the flexible and safe region identification for corresponding data generation, which is composed of flexible region identification in Section II-D1 and the safe region identification in Section II-D2 sequentially.

1) Flexible Region Identification: Based on [8], we formulate the flexible region identification problem for each load aggregator as the following optimization problem as follows:

$$d_{\text{ref}} = \min_{p_d} \left\| p_{\text{ref}} - \sum_{d\in\mathcal{LA}} p_d \right\| \tag{4a}$$

$$\text{s.t. } p_d \in \Omega_d, d \in \mathcal{D} \tag{4b}$$

where $\mathcal{D}$ in (4a) is the set of flexible resources, and Ω_d is the operation constraints of the flexible resource d from (1). Equation (4a) is to find whether p_{ref} could be disaggregated into a set of $p_d, d \in \mathcal{D}$ in a feasible way. For a given reference aggregated power p_{ref}, when the objective value d_{ref} of (4) is 0, the aggregate power p_{ref} is a feasible solution, denoted by $\mathbb{I}_{\text{flex}}[d_{\text{ref}} = 0]$ with $\mathbb{I}_{\text{flex}}[\cdot]$ as the flexibility indicator function. Based on (4), we generate the flexible region dataset $\mathcal{D}_{\text{flex}} = \{(p_{\text{ref}}, \mathbb{I}_{\text{flex}})\}$ by sampling massive p_{ref}, which provides the basis for flexible classification model training.

2) Safe Region Identification: We formulate the safe region identification problem as follows:

$$\text{Output: } \mathbb{I}_{\text{safe}}[W_s s \le z] \tag{5a}$$

$$\text{s.t. } s = A_s^p p_{\text{inj}} + A_s^q q_{\text{inj}} + b_s \tag{5b}$$

where $\mathbb{I}_{\text{safe}}[\cdot]$ is the safety indicator function; when the indicator value of (5) is 0, the power injection p_{inj} of distribution network lead to a safe operation state. Based on (5), we generate the safe region dataset $\mathcal{D}_{\text{safe}} = \{(p_{\text{inj}}, \mathbb{I}_{\text{safe}})\}$ by sampling massive p_{inj}, which provides the basis for the safe classification model training.

For concrete computation steps, given a sampled bus injection point $(p_{\text{inj}}, q_{\text{inj}})$: 1) compute the corresponding network state s (e.g., voltages and branch flows) using the adopted power-flow/security model, i.e., the mapping in (5b): $s = A_s^p p_{\text{inj}} + A_s^q q_{\text{inj}} + b_s$; 2) evaluate the security constraints $W_s s \le z$ (voltage/thermal limits represented in compact form); and 3) assign the binary label as an indicator of set membership, i.e., $\mathbb{I}_{\text{safe}} = 1$ if $W_s s \le z$ holds (safe) and $\mathbb{I}_{\text{safe}} = 0$ otherwise (unsafe). Repeating this procedure over many sampled points produces the labeled dataset $\mathcal{D}_{\text{safe}} = \{(p_{\text{inj}}, \mathbb{I}_{\text{safe}})\}$ used to train the safe region classifier.

It should be noted that $\mathcal{S}_{\text{flex}}$ and $\mathcal{S}_{\text{safe}}$ are accurate flexible and safe regions; $\mathcal{D}_{\text{flex}}$ and $\mathcal{D}_{\text{safe}}$ are corresponding dataset with features and labels.

III. Methodology

Based on the above, this section proposes to encode the data-driven flexible and safe regions for facilitating distribution network optimization, which is organized from the perspectives of advanced NNs formulation for classification in Section III-A and the equivalent learning model transformation in Section III-B.

A. Advanced Neural Network Model as Classifier

Beyond the conventional learning models, we further introduce the advanced NN models for safe and flexible region identification tasks, which feature high representative capability but low interpretability and are taken as advanced region classifiers. The main differences between the NN-based regression and classification are the output layers and loss function design. For completeness, the NN-based regression is introduced, and, on top of that, the corresponding NN-based classification is proposed sequentially.

1) NN-Based Regression:

a) Structure: We formulate a NN-based regression model $y = f_\theta^{\text{reg}}(x)$, parameterized by θ under the ReLU activation function in a fully connected structure, which takes the feature vector x as input and outputs continuous y as follows:

$$v_i^l = \text{ReLU}\left(\theta_{i0}^l + \sum_{j\in\mathcal{N}^{l-1}} \theta_{ij}^l v_j^{l-1}\right), l \in \mathcal{L}, i \in \mathcal{N}_l \tag{6a}$$

$$y = v^L = \theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1}, x = v^0 \tag{6b}$$

$$\text{where ReLU}(x) = \max(0, x) \tag{6c}$$

where v_i^l is the hidden ith neuron of layer l; θ_{i0}^l and θ_{ij}^l are linear transformation matrix between layer i and j; $\mathcal{L}$ is the set of layers; $\mathcal{N}_l$ is the set of neurons in layer l. Equation (6a) formulates the hidden layers function mapping, and (6b) formulates the output layer mapping.

b) Loss Function: Mean square error (MSE) is utilized as the loss function $\mathcal{L}_{\text{mse}}$ in (7) for regression model training under the stochastic gradient descent (SGD) algorithm

$$\mathcal{L}_{\text{mse}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{7}$$

where y_i and $\hat{y}_i$ are the true and predicted values.

2) NN-Based Classification:

c) Structure: On top of the NN-based regression of (6), we replace the output layer (6b) with sigmoid$(\cdot)$ to formulate the NN-based classification model $y = f_\theta^{\text{cla}}(x)$ parameterized by θ, which achieves the identification of the flexible and safe regions for distribution network operation, as follows:

$$v_i^l = \text{ReLU}\left(\theta_{i0}^l + \sum_{j\in\mathcal{N}^{l-1}} \theta_{ij}^l v_j^{l-1}\right), \quad l \in \mathcal{L}, i \in \mathcal{N}_l \tag{8a}$$

$$v^L = \text{sigmoid}\left(\theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1}\right) \tag{8b}$$

$$x = v^0, y = \mathbb{I}[v^L \ge \tau] \tag{8c}$$

$$\text{where ReLU}(x) = \max(0, x), \text{sigmoid}(x) = \frac{1}{1+e^{-x}} \tag{8d}$$

where $\mathbb{I}[\cdot]$ refers to the indicator function with τ as the classifier threshold. Based on replacing (6b) by the (8b), the indicator

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

Algorithm 1: Sparse Stochastic Gradient Descent Training Algorithm for Classification.

Input: Flexible and safe region datasets $\mathcal{D}_{flex}$ $\mathcal{D}_{safe}$, prediction model: $f_\theta(\cdot)$, sparse distributing: $\mathbb{S} = \{s^1, .., s^L\}$, updating scheduling: ΔT, T_{end}, α, f_{decay};
for $t = 1, \ldots, T$ **do**
  Calculate cross-entropy loss $L_{ce,t} = \sum L_{ce}(f_\theta(x_i), y_i)$;
  if t mod ΔT is 0 and $t \leq T_{end}$ **then**
    for $l = 1, \ldots, L$ **do**
      Calculate dropping and activating number $k = f_{decay}(t; \alpha, T_{end})(1 - s^l)N^l$;
      Drop $ArgTopK(-|\theta^l|, k)$;
      Activate $ArgTopK(\nabla_{\theta^l} L_{ce,t}, k)$;
    end for
  else
    $\theta \leftarrow \theta - \alpha \Delta_\theta L_{ce,t}$
  end if
end for
Output: Trained SNN classifier $f_\theta(\cdot)$ with sparse θ for flexible and safe region learning.

function mapping is added as (8c) to identify the flexibility of p_{ref} and the safety of $(p_{\text{inj}}, q_{\text{inj}})$.

d) Loss Function: Cross-entropy (CE) loss $\mathcal{L}_{\text{ce}}$ is utilized as the loss function in (9) for classification models training under the SGD algorithm

$$\mathcal{L}_{\text{ce}} = -\left(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right) \tag{9}$$

where y is the true label and $\hat{y}$ is the predicted probability of the label. We learn the optimal parameters θ to minimize $\mathcal{L}_{\text{ce}}$ for NN-based classification.

Remark 2. (Output of Classification Models): The outputs of classifiers refer to the probability between 0 and 1, which could be utilized to identify whether p_{ref} is flexible or not and whether $(p_{\text{inj}}, q_{\text{inj}})$ is safe or not.

3) SGD Training: Based on the above, we utilize NN models to train the NN-based regression models of (6) and NN-based classification models of (8) in a uniform SGD way with $\mathcal{L}_{\text{mse}}$ and $\mathcal{L}_{\text{ce}}$, which could be enhanced for downstream optimization with NN sparsity. Based on sparse training designs of [24], we propose the sparse (S-SGD) algorithm with iterative connection growing and dropping for training the SNN classifiers in Algorithm 1.

So based on the above, we build the flexible region classifier in (10a) and the safe region classifier in (10b)

$$y_{\text{flex}} = f_\theta^{\text{flex}}(p_{\text{ref}}) \tag{10a}$$

$$y_{\text{safe}} = f_\theta^{\text{safe}}(p_{\text{inj}}, q_{\text{inj}}). \tag{10b}$$

The data-driven regions cannot be applied directly in distribution network operation, whose reason is that the typical learned classifiers (especially NNs) provide an implicit, nonconvex decision boundary and act as opaque predictor as shown in (8); they do not directly yield an explicit set of algebraic constraints that an optimization solver can enforce. As a result, they cannot be embedded into dispatch/OPF models "as-is" to guarantee feasibility during optimization.

Discussion: Since the classifiers learn physical characteristics (electrical parameters, topology, transmission principles) implicitly from the solver-labeled datasets $\mathcal{D}_{\text{flex}}$ and $\mathcal{D}_{\text{safe}}$, the results may be biased by data coverage. Remedies include physics-informed NNs to embed physical knowledge explicitly, and active or adversarial (e.g., distributionally robust) training to improve resilience to limited or noisy data.

B. Mixed-Integer Linear Constraints Transformation of NN Models for Regression and Classification

The structures of NN-based models with (6) and (8) are quite complex, which could not be utilized in distribution network operation directly. So this article leverages the big-M relaxation to transform the trained NN-based models into a series of mixed-integer linear constraints in a uniform way. The flexible region classifier of (10a) and safe region classifier of (10b) could be further encoded in distribution network operation for economic and safe management. Based on the similarity of NN-based regression and classification, the equivalent transformation is composed of regression transformation in Section III-B1 and classification transformation in Section III-B2.

1) Regression Model Transformation: The main difficulty of (6) lies in the nonlinear mapping of the ReLU activation function. The big-M relaxation method [19] is utilized to transform the ReLU by introducing the binary variables. The transformation of ReLU is applied to all the activation functions in the (6) to achieve the complete transformation of NNs in the following:

$$v_i^1 = \theta_{i0}^1 + \sum_{j \in \mathcal{X}} \theta_{ij}^l x_j, i \in \mathcal{N}_1 \tag{11a}$$

$$v_i^l \geq \theta_{i0}^l + \sum_{j \in \mathcal{N}^{l-1}} \theta_{ij}^l v_j^{l-1}, l \in \mathcal{L}, i \in \mathcal{N}_l \tag{11b}$$

$$v_i^l \leq \theta_{i0}^l + \sum_{j \in \mathcal{N}^{l-1}} \theta_{ij}^l v_j^{l-1} - M_{l,i}^L (1 - t_{l,i}), l \in \mathcal{L}, i \in \mathcal{N}_l \tag{11c}$$

$$v_i^l \leq M_{l,i}^H t_{l,i}, l \in \mathcal{L}, i \in \mathcal{N}_l \tag{11d}$$

$$v_i^l \geq 0, \quad t_{l,i} \in \{0, 1\}, l \in \mathcal{L}, i \in \mathcal{N}_l \tag{11e}$$

$$\hat{y}_m = \theta_{i0}^L + \sum_{j \in \mathcal{N}^{L-1}} \theta_{ij}^L v_j^{L-1}, \quad m \in \mathcal{N}_M, i \in \mathcal{N}_l \tag{11f}$$

where $\hat{y}_m$ is the output of NNs of neural m; $t_{l,i}$ is the binary variables for ReLU relaxation of layer l in neuron i; $M_{l,i}^L$ and $M_{l,i}^H$ are small and big values of big-M relaxation. As shown in the transformation of NN into constraints in (11), sparsity means some θ of (11) are zero, leading to the fewer variables and constraints with simpler hybrid optimization of (17). The bounds $M_{l,i}^L, M_{l,i}^H$ are set from neuronwise preactivation ranges obtained by interval bound propagation from the dataset sampling boundaries, with a conservative global $M = 10^5$ fallback [19]. For a fixed binary the constraints are exact; a smaller

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

valid M tightens the LP relaxation and shrinks the branch-and-bound tree, whereas an excessively large M degrades numerical conditioning, so M is kept as small as validity permits. The resulting effect on computational performance is quantified in Figs. 6 and 7.

2) Classification Model Transformation: The hidden layer structures of NN-based regression (6) and classification (8) are the same, so the corresponding transformation is also the same. Different from the regression model, the output sigmoid activation function of classification model $v^L \in (0,1)$ cannot be linearized and embedded into the optimization model directly. When τ is the predetermined probability low bound, y will be 1 when $\theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1} \geq \ln(\frac{\tau}{1-\tau})$ of (8). So the output can be binarized as follows:

$$y = \begin{cases} 1, & \text{if } \theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1} \geq \ln(\frac{\tau}{1-\tau}) \\ 0, & \text{else.} \end{cases} \tag{12}$$

For example, given the threshold τ as 0.5, the output y will be 1 when $\theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1} \geq 0$.

Based on the above, we formulate the equivalent mixed-integer linear constraints of (8) as follows:

$$\text{Hidden Layer Constraints: (11a)-(11e)} \tag{13a}$$

$$\text{Binary Output: } \theta_0^L + \sum_{j\in\mathcal{N}^{L-1}} \theta_j^L v_j^{L-1} \geq \ln\left(\frac{\tau}{1-\tau}\right). \tag{13b}$$

We further explain the logical roles of and (12) and (13): (8) constructs the original NN *classification model* used in the learning stage. (13) is introduced in the *transformation stage* to embed the trained classifier into MILP-based dispatch. We do *not* enforce inside the dispatch MILP. Instead, the optimization model enforces the transformed constraint set: 1) the big-M linearization of the ReLU hidden layers (which determines the values of v^l), together with 2) the output inequality in (8).

Based on (13), we can transform both the trained flexible region classifiers and safe region classifier into the data-driven flexible region constraints $\mathcal{C}^{\text{flex}}(p_{\text{ref}})$ and safe region constraints $\mathcal{C}^{\text{safe}}(p_{\text{inj}})$ for distribution network operation application as follows:

$$y_{\text{flex}} = f_\theta^{\text{flex}}(p_{\text{ref}}) \xrightarrow[\text{via (13)}]{\text{MILP}} \mathcal{C}^{\text{flex}}(p_{\text{ref}}) \tag{14a}$$

$$y_{\text{safe}} = f_\theta^{\text{safe}}(p_{\text{inj}}, q_{\text{inj}}) \xrightarrow[\text{via (13)}]{\text{MILP}} \mathcal{C}^{\text{safe}}(p_{\text{inj}}, q_{\text{inj}}). \tag{14b}$$

IV. Encoding the Flexible and Safe Regions in Distribution Network Operation

The extracted learning-based flexible and safe regions could be encoded into the distribution network operation, which further facilitates the distribution network operation in a data-driven way. So this section firstly constructs the encoding framework for distribution network operation in Section IV-A, formulates the basic operation model in Section IV-B, encodes the learning-based flexible region for economic operation scenario with flexible region learning in Section IV-C, and encodes the learning-based flexible and safety region in safety operation scenario in Section IV-D,

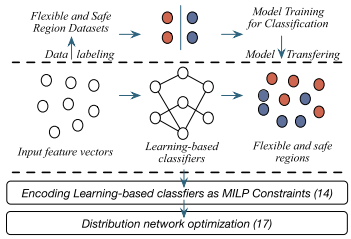


Fig. 1. Region encoding framework for distribution network operation.

A. Encoding Framework for Distribution Network Operation

Based on the learning-based flexible and safe regions, this article presents the region encoding framework for distribution network operation in Fig. 1.

For the classification in Fig. 1, we formulate the flexible and safe datasets, construct the flexible region and safe region classification models to learn from the corresponding dataset via various learning models, and evaluate the classification performance from multiple perspectives; for the optimization in Fig. 1, we transform the trained classification models to formulate the flexible and safe regions in the way of constraints to facilitate the distribution network operation.

The learned regions are data-driven and system-specific, with physical characteristics (parameters/topology) implicitly reflected in the offline solver-labeled datasets $\mathcal{D}_{\text{flex}}$ and $\mathcal{D}_{\text{safe}}$. When topology or key parameters change, accuracy is maintained by regenerating samples, relabeling, and retraining (or warm-start fine-tuning) the classifiers, implemented as periodic or event-triggered updates; for frequent switching, configuration indicators (e.g., switch status) can be added as inputs to learn a unified model across topologies.

B. Basic Distribution Network Operation Model

Based on the above framework, we first formulate the basic distribution network operation models to minimize the total operation cost under the system constraints as follows:

$$\min \sum_{t\in\mathcal{T}} \left\{ \sigma_{S,t}^P p_t^S + \sum_{i\in\mathcal{DER}} \sigma_{G,i}^P p_{i,t}^G \right\} \tag{15a}$$

$$\text{s.t.} \quad \sum_{jk\in\mathcal{E}} p_{jk,t} - \sum_{ij\in\mathcal{E}} (p_{ij,t} - r_{ij} l_{ij,t}) = p_{j,t}^G - p_{j,t}^D \; \forall ij \in \mathcal{E}, i \in \mathcal{N}_B \tag{15b}$$

$$\sum_{jk\in\mathcal{E}} q_{jk,t} - \sum_{ij\in\mathcal{E}} (q_{ij,t} - x_{ij} l_{ij,t}) = q_{j,t}^G - q_{j,t}^D \; \forall ij \in \mathcal{E}, i \in \mathcal{N}_B \tag{15c}$$

$$v_{j,t} = v_{i,t} + (r_{ij}^2 + x_{ij}^2) l_{ij,t} - 2(r_{ij} p_{ij,t} + x_{ij} q_{ij,t}) \quad \forall ij \in \mathcal{E} \tag{15d}$$

$$||2p_{ij,t} \quad 2q_{ij,t} \quad l_{ij,t} - v_{i,t}|| \leq l_{ij,t} + v_{i,t} \tag{15e}$$

$$(V_{i,t}^{\min})^2 \leq v_{i,t} \leq (V_{i,t}^{\max})^2 \; \forall i \in \mathcal{N}_B \tag{15f}$$

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

$$(I_{ij,t}^{\min})^2 \le l_{ij,t} \le (I_{ij,t}^{\max})^2 \; \forall ij \in \mathcal{E} \tag{15g}$$

$$0 \le p_{i,t}^G \le \bar{p}_i^G \; \forall i \in \mathcal{G} \; \forall t \in \mathcal{T} \tag{15h}$$

$$0 \le q_{i,t}^G \le \bar{q}_i^G \; \forall i \in \mathcal{G} \; \forall t \in \mathcal{T} \tag{15i}$$

Where $p_{j,t}^S$ and $q_{j,t}^S$ are the active and reactive power supply at substation; $p_{j,t}^G$ and $q_{j,t}^G$ are the active and reactive power supply of distributed energy resources; $p_{j,t}^D$ and $q_{j,t}^D$ are the active and reactive bus load demand; $v_{i,t}$ and $l_{ij,t}$ are the square of voltage magnitude and current; $\sigma_{S,t}^P$ is the active power price of the substation; $\sigma_{G,i}^P$ is the active power cost of DERs; $p_{ij,t}$ and $q_{ij,t}$ are the active and reactive power of branches; $\mathcal{T}$ and $\mathcal{E}$ are the set of operation periods and network branches; r_{ij} and x_{ij} are the resistance and reactance of branch ij.

Concretely, (15a) constructs the objective function of minimizing the operation cost in the distribution network with; (15b) and (15c) describe the bus active and reactive power balance with $p_{jk,t}$ and $q_{jk,t}$ as active and reactive power flow; (15d) denotes the branch voltage relationship with $v_{i,t}$ as the square of bus voltage; (15e) achieve the second-order cone relaxation of branch flow and bus voltage; (15f) and (15g) limit the bus voltage and branch power flow for safety consideration.

C. Learning-Based Economic Operation Model With Flexible Region Learning

Based on the above economic operation model, we further formulate the learning-based economic operation model by further considering the learning-based flexible regions for load aggregators, which includes the flexible constraints in the basic operation models of (15), as presented in the following:

$$\min \sum_{t\in\mathcal{T}} \left\{ \sigma_{S,t}^P p_t^S + \sum_{i\in\mathcal{DER}} \sigma_{G,i}^P p_{i,t}^G \right\} \tag{16a}$$

$$\text{s.t.} \quad \sum_{jk\in\mathcal{E}} p_{jk,t} - \sum_{ij\in\mathcal{E}} (p_{ij,t} - r_{ij} l_{ij,t}) = p_{j,t}^G - p_{j,t}^D - p_{j,t}^{\text{flex}} \tag{16b}$$

$$p_j^{\text{flex}} \in \mathcal{C}_j^{\text{flex}} \tag{16c}$$

$$(15c)-(15g) \text{ from basic model.} \tag{16d}$$

D. Learning-Based Safety Operation Model With Safe and Flexible Region Learning

Based on the learning-based flexible operation model, we further formulate the learning-based safety operation model by considering the both learning-based flexible and safe regions for load aggregators and distribution system operator, which includes the flexible constraints in the basic operation models of (15) and replaces the explicit operation constraints of (15) by learning-based safety region in the following:

$$\min \sum_{t\in\mathcal{T}} \left\{ \sigma_{S,t}^P p_t^S + \sum_{i\in\text{DER}} \sigma_{G,i}^P p_{i,t}^G \right\} \tag{17a}$$

$$\text{s.t.} p_j^{\text{inj}} = p_{j,t}^G - p_{j,t}^D - p_{j,t}^{\text{flex}}, q_{j,t}^{\text{inj}} = q_{j,t}^G - q_{j,t}^D \tag{17b}$$

$$p_j^{\text{flex}} \in \mathcal{C}_j^{\text{flex}}, (p_{j,t}^{\text{inj}}, q_{j,t}^{\text{inj}}) \in \mathcal{C}^{\text{safe}} \tag{17c}$$

$$s0 \le p_{i,t}^G \le \bar{p}_i^G \; \forall i \in \mathcal{G} \; \forall t \in \mathcal{T} \tag{17d}$$

$$s0 \le q_{i,t}^G \le \bar{q}_i^G \; \forall i \in \mathcal{G} \; \forall t \in \mathcal{T}. \tag{17e}$$

The above model corresponds to the *safety-operation* variant where we intentionally *replace* the explicit network security constraints in (15) (including the branch-flow/voltage feasibility conditions) by the learned safe region constraint $(p_{j,t}^{\text{inj}}, q_{j,t}^{\text{inj}}) \in \mathcal{C}^{\text{safe}}$, where (15c) is omitted.

All the above basic, learning-based economic, and learning-based safety operation models are finally formulated in mixed-integer linear programming (MILP) mathematically, which can be solved by the off-the-shelf solvers effectively. s From the DSO perspective, flexible-region constraints ensure the scheduled flexible power is implementable by the aggregator portfolio, while network (or learned safe-region) constraints impose the security limits; with multiple aggregators, the optimizer exploits the spatial and temporal diversity of flexibility to cut cost while avoiding local voltage/thermal issues.

V. Case Study

The case study utilizes the modified 33-bus and 123-bus systems with various distributed energy resources to verify the effectiveness of learning the safe and flexible regions via fusing classification and optimization. The proposed hybrid data-driven models in Section V are composed of the learning stage of the classification models and the optimization stage of the distribution network operation. So, for each stage, we construct the classification models to learn safe and flexible regions with the Sklearn package [25] for conventional learning models, Pytorch package [26] for NNs and formulate the distribution network operation models with the Cvxpy package [27] equipped by Gurobi 9.0. All the experiments are coded by Python 3.10 and implemented in a MacBook Pro laptop with RAM of 16 GB, CPU Intel Core i7 (2.6 GHz). For region learning, the NNs use 100 hidden neurons trained with Adam; each load aggregator comprises 50 EVs and 50 energy-storage units, and case k denotes k photovoltaic (PV) units ($k = 1, \ldots, 4$). The 33-bus system uses two aggregators (buses 3 and 10) and the 123-bus system four (buses 5, 20, 70, 80), and the big-M value is 10^5.

A. Data Preparation

1) Data Sampling Strategy: We utilize the region identification strategies in Section II-D to generate the proper dataset for model learning with the flexible region identification of (4) and the safe region identification of (5).

We firstly construct the sampling boundaries for flexible and safe regions For flexible region boundary, we conduct Minkowski sum on each upper and lower bounds of each distributed load for flexible regions by (18); for safe region boundary, the minimum and maximum active/reactive power in each node is determined by its operation history. *The data outside the boundaries are all infeasible and unsafe, and the data within*

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

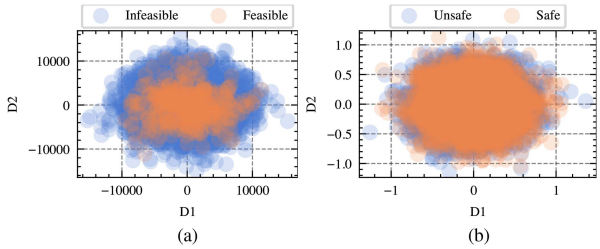


Fig. 2. PCA analysis in flexible and safe regions. (a) PCA analysis in a flexible region. (b) PCA analysis in a safe region.

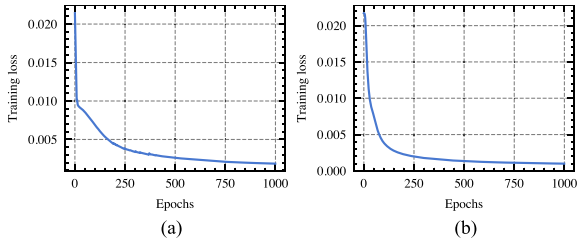


Fig. 3. Training process of the flexible and safe regions learning. (a) Training process in the flexible region learning. (b) Training process in the safe region learning.

the boundaries could be feasible/infeasible and safe/unsafe data

$$P_{\mathrm{ub},t} = \sum_{d\in\mathcal{D}} p_{d,t,\max} \ \forall t \in \mathcal{T} \tag{18a}$$

$$P_{\mathrm{lb},t} = \sum_{d\in\mathcal{D}} p_{d,t,\min} \ \forall t \in \mathcal{T} \tag{18b}$$

$$p_{d,t,\max}, p_{d,t,\min} \in (1). \tag{18c}$$

On top of the above boundaries, we generate candidate operating points by independent uniform sampling within prescribed boundaries as follows:

$$p_{\mathrm{ref}} = U(P_{\mathrm{lb}}, P_{\mathrm{ub}}) \tag{19a}$$

$$(p_{\mathrm{inj}}, q_{\mathrm{inj}}) = \left(U\left(P^{\mathrm{inj}}_{\min}, P^{\mathrm{inj}}_{\max}\right), U\left(Q^{\mathrm{inj}}_{\min}, Q^{\mathrm{inj}}_{\max}\right)\right). \tag{19b}$$

Based on (19) sampling, the flexible power $p_{\mathrm{ref}} \in \mathbb{R}^T$ and distribution network power injections $(p_{\mathrm{inj}}, q_{\mathrm{inj}}) \in \mathbb{R}^{2N_B}$ are delivered to (4) and (5) for the identification of flexible and safe data. After that, the flexible dataset $\mathcal{D}_{\mathrm{flex}}$ for each load aggregator and safe dataset $\mathcal{D}_{\mathrm{safe}}$ are constructed for flexible and safe region classifiers of (10) to learn to identify the safe and flexible regions. The numbers of flexible and safe region datasets are both set as 10 000.

We note that the data distribution in $\mathcal{D}_{\mathrm{flex}}$ and $\mathcal{D}_{\mathrm{safe}}$ is determined by (4) and (5), which could not formulated in exactly analytical expression. So we could not sample around the possible feasible boundaries for data coverage and convergence enhancing.

B. Learning Performance Analysis

Based on the above model, this part analyzes the learning performance of different learning models in identifying the safe and flexible regions in the systems.

1) Dimension Reduction Analysis of Safe and Flexible Regions: Based on the data sampling of (19), we formulate the safe dataset $\mathcal{D}_{\mathrm{safe}}$ and flexible dataset $\mathcal{D}_{\mathrm{flex}}$, where the input p_{ref} and p_{inj} are high-dimensional. For the clear distribution of feasible/infeasible and safe/unsafe data, the principal component analysis (PCA) [28] is leveraged to achieve dimension-reduction for visualization by transforming the p_{ref} and p_{inj} into 2-D vectors, where the 2-D vectors are scattered in different colors for feasibility and safety analysis, as shown in Fig. 2.

2) Training Process Analysis of Region Learning: After visualization analysis, we utilize the SNNs to learn how to classify the safe and flexible regions in the systems, as shown in Fig. 3. The training processes of both the flexible and safe regions converge after 500 epochs, with the cross-entropy loss decreasing fast at first and flattening later, verifying the convergence and effectiveness of the proposed region learning for the flexible and safe regions.

In the training stage, the results demonstrate that the proposed SNN classifiers can efficiently and accurately learn the boundaries of flexible and safe regions. The rapid convergence and low final loss values confirm the suitability of the approach for practical distribution network operation applications.

3) Evaluation of Classification Models:

a) Numerical Evaluation: The following Tables II and III analyze the learning performance of flexible and safe regions from the metrics perspective of accuracy, recall, F1, precision, and sparsity. Due to the embedding of learning models in optimization, we further analyze the parameters and sparsity of different learning models for learning and calculation analysis. This shows that enforcing the data-driven *flexible-region* constraints characterizes the aggregator's feasible set of aggregated power and helps unlock its regulation potential in a model-free manner, without requiring practitioners to know detailed distributed energy resource parameters.

For flexible region learning in Table II, SNN attains classification performance comparable to the fully connected neural network (FCNN): a higher precision (0.976 versus 0.908) with comparable recall and F1 (0.868 and 0.908), while FCNN attains a marginally higher accuracy (0.916 versus 0.904). Both SNN and FCNN demonstrate significant learning performance enhancement than conventional LR and support vector classifiers (SVC), due to the learning capability with more parameters. For parameters and sparsity, conventional learning models feature the fewest parameters with no sparsity. FCNN features the most parameters with 6801 parameters with no sparsity, and the SNN features 681 parameters with a 10.0% sparsity rate. s This highlights the potential of enforcing data-driven *safe region* constraints to restrict bus-injection decisions to operating points that satisfy network security limits, enabling model-driven safety enforcement without requiring practitioners to have detailed distribution-network parameters.

For safe region learning in Table III, FCNN achieves slightly better classification performance than SNN with the same accuracy, but higher precision, recall, and F1 score in the training dataset. Similarly, both SNN and FCNN demonstrate significant learning performance enhancement than conventional LR and SVC. Similar to flexible region learning, for parameters and sparsity, conventional learning models feature the fewest

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE II
FLEXIBLE REGION IDENTIFICATION ANALYSIS OF DIFFERENT MODELS

Model	Accuracy	Precision	Recall	F1	Parameter Number	Sparsity Rate
LR	0.5416	0.8771	0.5416	0.6301	24	0%
SVC	0.8172	0.904	0.8584	0.904	24	0%
FCNN	0.91649	0.908	0.8679	0.908	6801	0%
SNN	**0.904**	**0.9763**	**0.8679**	**0.908**	**681**	**10.0%**

It should be noted LR refers to the logistic regression; SVC refers to the support vector classifier; FCNN refers to the fully connected neural network; SNN refers to the SNN; the sparsity rate is calculated by the ratio of parameters of the proposed SNN to the parameters of full parameters.

TABLE III
SAFE REGION IDENTIFICATION ANALYSIS OF DIFFERENT MODELS

Model	Accuracy	Precision	Recall	F1	Parameter Number	Sparsity Rate
LR	0.5389	0.5385	0.5389	0.5357	66	0%
SVC	0.5657	0.5686	0.5657	0.5563	66	0%
FCNN	1.0	1.0	1.0	1.0	2801	0%
SNN	**1.0**	**0.9948**	**0.9948**	**0.9948**	**281**	**10.0%**

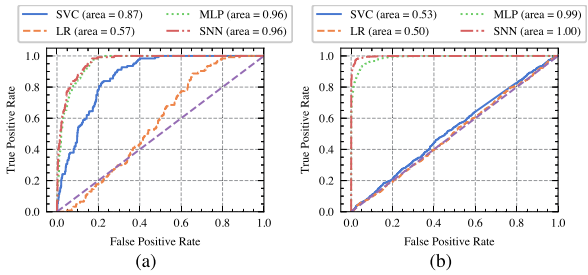


Fig. 4. Region identification performance of different learning models in the testing set. (a) Flexible region identification performance of different learning models in the testing set. (b) Safe region identification performance of different learning models in the testing set.

parameters with 66 parameters with no sparsity. FCNN achieves the most parameters with 2801 parameters with no sparsity, and the proposed SNN achieves 281 parameters with a 10.0% sparsity rate

b) ROC Curve Analysis: Based on the above metrics, we further analyze the learning performance based on the receiver operating characteristic (ROC) curve in the testing dataset in the following Fig. 4(a) and (b). ROC is composed of the x-axis with true positive rate (TPR) and the y-axis with false positive rate (FPR) for evaluating the classification models. TPR refers to the ratio of true positives to the sum of true positives and false negatives, and FPR refers to the ratio of false positives to the sum of false positives and true negatives. The ROC curve formulates the relation between TPR and FPR at different classification thresholds. Lowering the classification threshold of (8c) could classify more items as positive, thus increasing both False Positives and True Positives. The area under the ROC curve (AUC) refers to the entire 2-D area underneath the entire ROC curve, where AUC provides an aggregate measure of performance across all possible classification thresholds of (8c). Higher AUC means higher classification performance.

For the flexible region learning evaluation in the testing dataset, as shown in Fig. 4(a), the proposed SNNs feature the largest AUC with 0.97, which is similar to the MLP with 0.97 and much higher than LR with 0.57 and SVC with 0.87, demonstrating higher learning capability of advanced NNs as classifiers than conventional learning models as classifiers in learning the flexible region. For the safe region learning evaluation in the testing dataset, as shown in Fig. 4(b), the proposed SNNs feature the largest AUC with 1.00, which is comparable with the MLP with 0.99 and further higher than LR with 0.50 and SVC with 0.53, demonstrating higher learning capability of advanced NNs as classifiers than conventional learning models as classifiers in learning the safe region.

In the classification evaluation stage, the experiments demonstrate that SNN classifiers achieve superior performance in identifying flexible and safe regions compared to conventional models, such as LR and SVC. SNNs provide higher accuracy, precision, recall, and F1 scores while maintaining significantly fewer parameters, resulting in improved computational efficiency. The ROC curve analysis further confirms the robustness of SNNs, with AUC values close to 1.0, indicating excellent classification capability. These results validate the effectiveness of the proposed approach for practical distribution network operation, ensuring both economic and safe management with scalable and efficient optimization. Tree-ensemble classifiers are likewise MILP-encodable but enlarge the embedded model, while kernel classifiers are not directly representable as sparse ReLU constraints; SNN remains the most favorable in jointly balancing accuracy, sparsity, and embedding tractability.

C. Application Performance of Classification Models Under Various Scenarios

Based on the learning performance analysis, this part further analyzes the application of classification models in distribution network operation. We analyze economic operation of the hybrid model from the following economic metrics 1) *Fixed cost* (baseline) corresponds to the dispatch with fixed demand (without enforcing aggregator flexibility realizability); 2) *Flexible cost* corresponds to dispatch with the learned flexible-region constraint enforced (and network safety ensured by explicit network constraints in the economic-analysis setting); 3) *Safe cost* corresponds to dispatch with both learned flexible-region and learned safe region constraints enforced (in the safety-analysis setting); and 4) *Cost enhancement* is computed as the relative change of total operating cost with respect to the stated baseline. The

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE IV
COST ANALYSIS WITH FLEXIBLE REGION UNDER DIFFERENT PV PENETRATION

Cases	Flexible Cost($)	Fixed Cost($)	Enhanced
Case 1	667.17	807.61	17.39%
Case 2	612.63	753.59	18.71%
Case 3	567.17	708.98	20.00%
Case 4	538.76	679.53	20.71%

It should be noted flexible cost refers to the distribution network operation cost with flexible region; fixed cost refers to the distribution cost with fixed load; enhanced refers to the cost reduction compared to the fixed demand model.

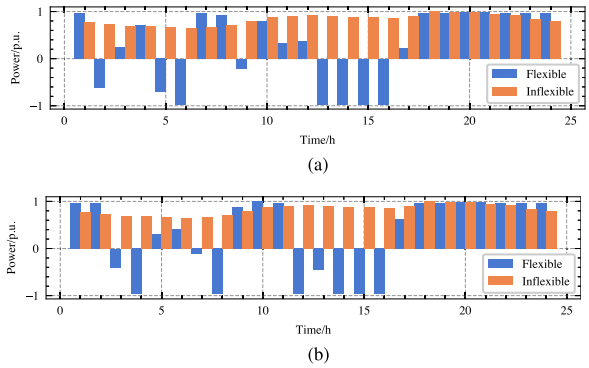


Fig. 5. Temporal aggregate power dispatch results of two representative load aggregators for case 4 in normalized p.u. values. (a) Temporal load demand of load aggregator 1 under flexible and inflexible load demand. (b) Temporal load demand of load aggregator 2 under flexible and inflexible load demand.

above metrics could be possible options for real-world operation evaluation.

1) Economic Operation Analysis: We analyze the economic value of the flexible region in distribution network operation by considering only the flexible region and setting the explicit safety region constraints, as shown in Tab. IV. With the increase of PV penetration, the system operation cost with the flexible region is lower than that with fixed demand. The *economic metric* in this part is the optimal objective value of the distribution network operation problem, i.e., the total operation cost over the scheduling horizon in (16) and (17). We also clarify the meaning of "cost enhancement" as the relative change of the total operation cost with respect to the corresponding baseline case. "Enhanced" quantifies the relative cost reduction achieved by enabling flexibility, computed as Enhanced $= \frac{\text{Fixed Cost}-\text{Flexible Cost}}{\text{Fixed Cost}} \times 100\%$.

Fig. 5 visualizes the temporal aggregate power schedules of two representative load aggregators (see Case 4) in normalized p.u. values, highlighting how the aggregator-level demand can be reshaped over time when flexibility is enabled (i.e., when the learned flexible-region constraint is enforced).

The above figure shows that enforcing the flexible-region constraint enlarges the feasible dispatch set by allowing *realizable* demand shifting (consistent with EV/ES portfolio limits), which enables the DSO to find lower-cost operating points while still satisfying the explicit network/security constraints in the economic-operation setting. We also clarify that different aggregators can exhibit different flexibility patterns, illustrating heterogeneity that is captured by the proposed region-encoding framework.

TABLE V
COST ANALYSIS WITH FLEXIBLE AND SAFE REGIONS UNDER DIFFERENT PV PENETRATIONS

Cases	Flexible Cost($)	Safe Cost ($)	Violation
Case 1	667.17	667.17	0%
Case 2	612.63	612.63	0%
Case 3	567.17	567.17	0%
Case 4	538.76	538.76	0%

It should be noted violation denotes the constraint violation rates.

2) Safety Operation Analysis: Based on the economic operation analysis, we further verify the effectiveness of the learning-based safety region by replacing explicit safety region constraints. Table V shows the operation cost with learning-based flexible and safe regions. Compared to only the learning-based flexible region, the consideration of both learning-based flexible and safe regions features the same operation economy with no cost addition. The learned safe-region constraint yields 0% *empirical* constraint violation across the tested scenarios. We emphasize that the learned region is a data-driven approximation of the security set rather than a hard feasibility guarantee; its conservativeness can be tuned through the classifier threshold τ (a larger τ shrinks the learned region, trading a small cost increase for fewer violations), while its worst-case feasibility under large out-of-distribution shifts is not guaranteed and is left to future scenario-aware/robust region learning. The flexible cost and safe cost in Table V in each scenario are the same, demonstrating that the learned safe-region constraint closely matches the explicit safety-feasible set in the tested scenarios.

As the number of PV units increases, the total operation cost decreases. The primary reason is that PV generation supplies a larger portion of the local demand, reducing the net power purchased from the main grid (which is a dominant cost component in the objective). Higher PV penetration can in general increase the risk of operational violations (e.g., overvoltage or branch thermal overload) due to reverse power flow and more variable injections. In our framework, these risks are addressed by explicitly enforcing network security through: (i) the explicit safety constraints in the economic analysis, and (ii) the learning-based safe region constraints in the safety-aware analysis. As a result, the reported violation rates remain 0% across the considered PV-penetration cases in the presented study, demonstrating that the dispatch decisions stay within the safe operating set for the tested scenarios.

PV uncertainty affects both regions: variable net injections can push operating points toward the safe-region boundary (more likely under high penetration), while the *usable* aggregator flexibility may shrink as more is reserved to counter renewable ramps. Within our framework this is handled by (i) receding-horizon reoptimization with updated forecasts while keeping the learned regions fixed, and/or (ii) scenario-aware region learning that labels data under multiple PV realizations to obtain a more conservative region. Extreme events (e.g., abrupt PV loss and equipment failure) induce distribution shift and are not covered by the present scenario set, motivating scenario-aware. and robust region learning as future work.

3) Calculation Efficiency Analysis: Sparsity designs of NNs can not only enhance the inference speed but also accelerate

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

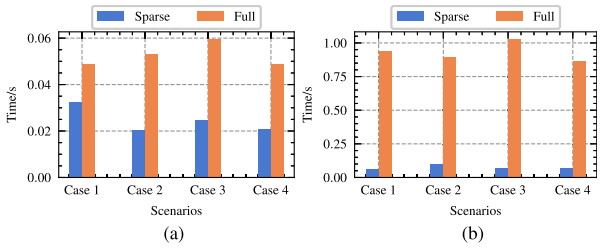


Fig. 6. Calculation time comparison with SNN and FCNN under various scenarios. (a) Calculation time comparison with flexible learning. (b) Calculation time comparison with flexible and safe learning.

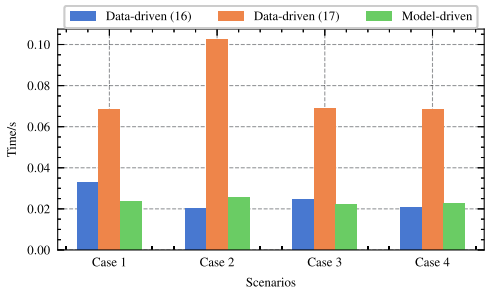


Fig. 7. Computation time comparison of the proposed data-driven constraint-learning formulation and conventional model-based formulation.

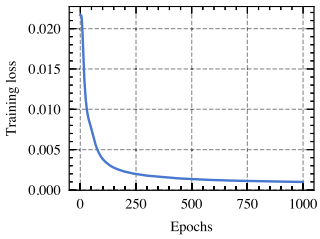


Fig. 8. Safe region training performance analysis in the 123-bus system.

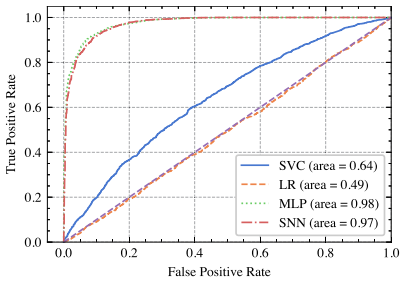


Fig. 9. Safe region identification performance of different learning models in the testing set in the 123-bus system.

the optimization of embedded neural networks. Fig. 6 compares the calculation speed for various cases. Obviously, the sparse design achieves around 2.13 times acceleration in the economic scenario of Fig. 6(a) and around 12.18 times acceleration in the safety scenario of Fig. 6(b). The above results demonstrate the computational efficiency of SNN with fewer optimization variables and reduced problem complexity. We note that sparsification itself is a well-established technique and not an original contribution of this work. Our focus is on demonstrating how

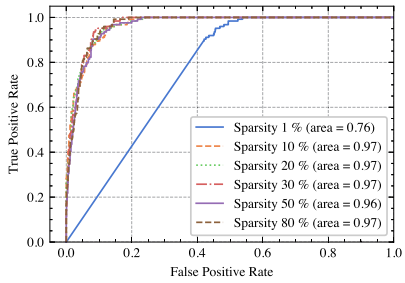


Fig. 10. Flexible region identification comparison under different sparsity ratios.

SNNs, when combined with constraint learning, can significantly learn the safe and flexible regions accurately and further accelerate mixed-integer optimization for distribution network operation compared to conventional dense models due to the sparse design.

In the optimization stage, the experiments confirm that encoding data-driven flexible and safe regions using SNNs enables accurate, efficient, and scalable distribution network operation. SNNs outperform conventional models in classification metrics while requiring fewer parameters, and both flexible and safe regions are identified with near-perfect ROC–AUC values. Incorporating flexible region constraints reduces operational costs by up to 20% as PV penetration increases, and learning-based safe region constraints achieve 0% empirical violation in the tested scenarios, supporting secure operation. Sparse models also accelerate optimization by up to 12 times, and the approach maintains high performance when scaled to larger systems, demonstrating its practical applicability for economic and secure network management.

We have compared computation time comparison of the proposed data-driven constraint-learning formulation and conventional model-based formulation in the following Fig. 7. The proposed data-driven constraint learning formulation of (16) shows the similar calculation to conventional model-based formulation, and (17) shows higher calculation time than conventional formulation due to the complex transformation of NNs.

Compared with the conventional model-based formulation, the key innovation of the proposed data-driven constraint-learning formulation lies in that it does not require massive flexible resource model parameters in (1) and the detailed system parameters in (15) with privacy preserving benefits.

D. Scalability Analysis

For scalability analysis, we apply the proposed encoding approach to the 123-bus distribution network system. The safe region classifier is retrained specifically for the 123-bus system based on Algorithm 1. The training process of safe region identification is shown in Fig. 8, demonstrating the convergence effectiveness of the proposed approach.

The safe region learning for case 123 is shown in Fig. 9. The SNN and MLP show similar ROC values, which are much higher than conventional machine learning models of LR and SVC. It demonstrates the scalability effectiveness of region learning in larger networks.

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.

This article has been accepted for inclusion in a future issue of this journal. Content is final as presented, with the exception of pagination.

TABLE VI
SAFE ANALYSIS OF DIFFERENT MODELS IN 123-BUS SYSTEMS

Cases	Economy ($) of (16)	Safety ($) of (17)
Cost	747.61	747.61
Violation	0%	0%
Solution Time	0.24832s	0.37993s

For the region learning in optimization, the corresponding operation result comparison is shown in Table VI. The proposed learning-based safety operation model of (17) shows the same cost and violations with (16) but features higher solution of 0.1316 s due to data-driven constraints, verifying its scalability.

E. Classification Analysis Under Different SNNs

Regarding the sparsity rate of the SNN, the 10% value was chosen empirically to balance model expressiveness and computational efficiency, and is consistent with common practice in the literature for SNNs. Fig. 10 presents the flexible region identification performance under different sparsity ratios. The identification performance with 10% to 80% was similar around 0.96, and the 10% feature lower parameters. When sparsity was set as 1%, the identification performance was reduced to 0.76. The above experiments demonstrate the rationality of 10% setting.

VI. CONCLUSION

This article fuses classification and optimization to encode the data-driven flexible and safe regions in the distribution network operation. The identification of data-driven flexible and safe regions is cast as binary classification tasks. Specifically, SNNs are utilized based on their classification performance for region learning and fewer parameters for network optimization. The trained region classifiers are transformed into a series of MILP constraints and encoded in the distribution network's economic and safe operation. The classification accuracy of region learning is verified via multiple metrics with higher prediction accuracy and fewer parameters, and the economy and empirical safety of data-driven regions are verified for the tested scenarios. Future work may consider the physics-informed NN, active/adversarial learning for learning the regions trustfully, extreme events learning, and imbalanced data processing problem in region data sampling, stronger tests under forecast errors and boundary (out-of-distribution) samples, and benchmarking against more competitive surrogate/constraint-learning baselines

REFERENCES

[1] M. Wussow et al., "Exploring the potential of non-residential solar to tackle energy injustice," *Nature Energy*, vol. 9, pp. 654–663, 2024. [Online]. Available: https://doi.org/10.1038/s41560-024-01485-y

[2] L. Xie et al., "Energy system digitization in the era of AI: A three-layered approach toward carbon neutrality," *Patterns*, vol. 3, no. 12, 2022, Art. no. 100640.

[3] L. Xie, X. Zheng, Y. Sun, T. Huang, and T. Bruton, "Massively digitized power grid: Opportunities and challenges of use-inspired AI," *Proc. IEEE*, vol. 111, no. 7, pp. 762–787, Jul. 2023.

[4] D. Yue, Z. He, and C. Dou, "Cloud-edge collaboration-based distribution network reconfiguration for voltage preventive control," *IEEE Trans. Ind. Informat.*, vol. 19, no. 12, pp. 11542–11552, Dec. 2023.

[5] N. Liu, C. Li, L. Chen, and J. Wang, "Hybrid data-driven and model-based distribution network reconfiguration with lossless model reduction," *IEEE Trans. Ind. Informat.*, vol. 18, no. 5, pp. 2943–2954, May 2022.

[6] S. Wang and W. Wu, "Aggregate flexibility of virtual power plants with temporal coupling constraints," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 5043–5051, Jun. 2021.

[7] Q. Li, J. Liu, B. Cui, W. Song, and J. Ye, "Distribution system flexibility characterization: A network-informed data-driven approach," *IEEE Trans. Smart Grid*, vol. 15, no. 1, pp. 1188–1191, Jan. 2023.

[8] S. Taheri, V. Kekatos, S. Veeramachaneni, and B. Zhang, "Data-driven modeling of aggregate flexibility under uncertain and non-convex device models," *IEEE Trans. Smart Grid*, vol. 13, no. 6, pp. 4572–4582, Jun. 2022.

[9] I. A. Sajjad, G. Chicco, and R. Napoli, "Definitions of demand flexibility for aggregate residential loads," *IEEE Trans. Smart Grid*, vol. 7, no. 6, pp. 2633–2643, Jun. 2016.

[10] S. Wang, W. Wu, C. Lin, and B. Chen, "A dynamic equivalent energy storage model of natural gas networks for joint optimal dispatch of electricity-gas systems," *IEEE Trans. Sustain. Energy*, vol. 15, no. 1, pp. 621–632, Jan. 2024.

[11] Y. Wen, Z. Hu, J. He, and Y. Guo, "Improved inner approximation for aggregating power flexibility in active distribution networks and its applications," *IEEE Trans. Smart Grid*, vol. 15, no. 4, pp. 3653–3665, Jul. 2023.

[12] X. Chen and N. Li, "Leveraging two-stage adaptive robust optimization for power flexibility aggregation," *IEEE Trans. Smart Grid*, vol. 12, no. 5, pp. 3954–3965, May 2021.

[13] P. V. Hentenryck, "Machine learning for optimal power flows," *INFORMS Tut. Operations Res.*, pp. 62–82, 2021, doi: 10.1287/educ.2021.0234.

[14] N. Zhang et al., "Data-driven security and stability rule in high renewable penetrated power system operation," *Proc. IEEE*, vol. 111, no. 7, pp. 788–805, 2023.

[15] G. Chen, H. Zhang, and Y. Song, "Efficient constraint learning for data-driven active distribution network operation," *IEEE Trans. Power Syst.*, vol. 39, no. 1, pp. 1472–1484, Jan. 2024.

[16] L. Sang, Y. Xu, and H. Sun, "Encoding carbon emission flow in energy management: A compact constraint learning approach," *IEEE Trans. Sustain. Energy*, vol. 15, no. 1, pp. 123–135, Jan. 2024.

[17] P. Li, W. Wu, X. Wang, and B. Xu, "A data-driven linear optimal power flow model for distribution networks," *IEEE Trans. Power Syst.*, vol. 38, no. 1, pp. 956–959, Jan. 2023.

[18] Y. Wang, J. Qiu, and Y. Tao, "Optimal power scheduling using data-driven carbon emission flow modelling for carbon intensity control," *IEEE Trans. Power Syst.*, vol. 37, no. 4, pp. 2894–2905, Apr. 2022.

[19] D. Maragno, H. Wiberg, D. Bertsimas, I. Birbil, D. den Hertog, and A. O. Fajemisin, "Mixed-integer optimization with constraint learning," *Oper. Res.*, vol. early access, 2023.

[20] G. Chen, H. Zhang, and Y. Song, "Efficient constraint learning for data-driven active distribution network operation," *IEEE Trans. Power Syst.*, vol. 39, no. 1, pp. 1472–1484, Jan. 2024.

[21] Y. Song, G. Chen, and H. Zhang, "Constraint learning-based optimal power dispatch for active distribution networks with extremely imbalanced data," *CSEE J. Power Energy Syst.*, vol. 10, no. 1, pp. 51–65, 2024.

[22] G. Chen, H. Zhang, and Y. Song, "Adversarial constraint learning for robust dispatch of distributed energy resources in distribution systems," *IEEE Trans. Sustain. Energy*, vol. 16, no. 2, pp. 1139–1152, Feb. 2025.

[23] F. L. Müller, J. Szabó, O. Sundström, and J. Lygeros, "Aggregation and disaggregation of energetic flexibility from distributed energy resources," *IEEE Trans. Smart Grid*, vol. 10, no. 2, pp. 1205–1214, Feb. 2019.

[24] J. Frankle and M. Carbin, "The lottery ticket hypothesis: Finding sparse, trainable neural networks," in *Proc. 7th Int. Conf. Learn. Represent*. New Orleans, LA, USA, May 6-9, 2019.

[25] F. Pedregosa et al., "Scikit-learn: Machine learning in python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[26] A. P. et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. 2019 Neural Inf. Process. Syst.*, 2019, pp. 8024–8035.

[27] S. Diamond and S. Boyd, "CVXPY: A python-embedded modeling language for convex optimization," *J. Mach. Learn. Res.*, vol. 17, no. 83, pp. 1–5, Jan. 2016.

[28] I. J. Goodfellow, Y. Bengio, A. Courville, and D. Learning. Cambridge, MA, USA: MIT Press, 2016. http://www.deeplearningbook.org

Authorized licensed use limited to: North China Electric Power University. Downloaded on September 13,2026 at 16:32:53 UTC from IEEE Xplore. Restrictions apply.